

RYAN KELLY

Principal Software Engineer

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 rtkelly13

Technologies

Overview

Creator of **Parquet.SourceGenerator** & **Parquet.TypeProvider** for zero-reflection high-throughput data processing.

Co-organizer of the **AWS Liverpool User Group** & technical speaker on functional cloud architectures.

Core & Languages

- .NET (C# / F# / Native AOT)
- TypeScript
- Python
- SQL

Cloud, Data & Infrastructure

- AWS
- Apache Iceberg / Parquet
- IaC (Pulumi / Terraform)
- Elasticsearch
- OpenTelemetry (OTEL)

Engineering & Practices

- Roslyn Source Generators
- Compiler Plugins / Type Providers
- CI/CD (GitHub Actions)
- Distributed Systems
- AI Code Agent Workflows
- TDD & High-Throughput Benchmarking

References

Robert Roe — Tech Lead at Zuto
Dan Cartwright — Product Manager at AutoCoding Systems
Contact Details: Provided on request

Personal Profile

Hands-on Principal Software Engineer specializing in scalable full-stack and cloud architectures, high-throughput data processing, and enterprise modernization. Proven track record architecting distributed systems processing billions of events, driving zero-downtime legacy migrations, and establishing modern design systems. Active technical leader, open-source contributor in high-performance .NET/Parquet tooling, and co-organizer of the AWS Liverpool user group.

Employment History

May 2019 - Present	Software Developer (Senior / Principal)	Sentric Music
	• Architected and engineered the core royalty processing system to scale reliably across billions of usage events and financial records. • Spearheaded enterprise Apache Iceberg adoption and Python data engineering tooling to modernize data lake warehousing. • Led technical strategy migrating legacy .NET Framework services to modern .NET, establishing a Storybook React design system backed by REST APIs. • Pioneered AI Code Agent workflows across engineering, optimizing developer experience and autonomous tooling adoption. • Introduced centralized observability via Serilog, Elastic Stack, and OpenTelemetry (OTEL) for real-time monitoring of event pipelines. • Standardized CI/CD on GitHub Actions and codified cloud infrastructure using Pulumi for AWS resource management.	
Nov 2017 - May 2019	Software Developer (Professional / Senior)	Zuto
	• Engineered high-availability cloud lender integrations on AWS, maintaining and modernizing scalable fintech services with strict observability standards.	
Jul 2014 - Nov 2017	Software Developer (Placement / Graduate)	AutoCoding Systems Ltd
	• Built packaging line automation software deployed to factory floors worldwide, developing critical features for the automated date and code printing engine.	

Education

2012 - 2016	BSc Computer Science with a Year in Industry	University of Liverpool
	First Class Honours Final year dissertation on 3D procedural maze generation with statistical graph analysis, built in Unity 3D with C#, SOLID, and TDD practices.	
2010 - 2012	A-Levels	Carmel College
	Computing (A), Chemistry (A), Mathematics (B)	